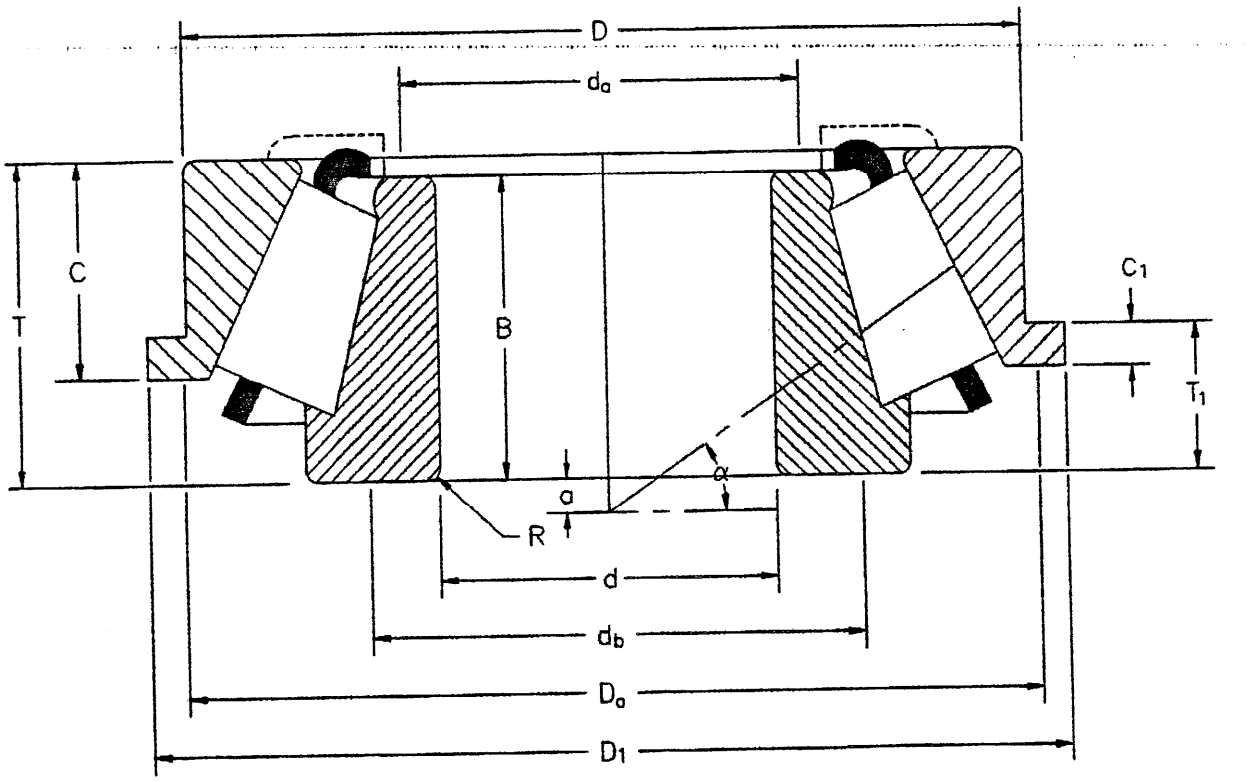
FEDERAL SPECIFICATION SHEET

BEARING, ROLLER, TAPERED, SINGLE ROW, STEEP ANGLE, FLANGED CUP (TYPE TSSF)

This specification is approved by the Administrator, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring the bearings described herein, shall consist of this document and the latest issue of FF-B-187/GEN.



BEARING CONFIGURATION

FSC 3110

AMSC N/A

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH	T1 STANDOUT	B CONE WIDTH	C CUP WIDTH	R 2/ SHAFT FILLET RADIUS	D1 FLANGE DIA	C1 FLANGE WIDTH	db BACKING SHOULDER DIAMETERS SHAFT	da BACKING SHOULDER DIAMETERS SHAFT	Da BACKING SHOULDER DIAMETERS HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) RADIAL	a 3/ EFFECTIVE LOAD CENTER
01 *	21075-21212-B	0.7500	2.1250	0.8750	0.4062	0.8598	0.6250	0.06	2.2772	0.1562	1.24	1.04	2.05	0.99	7500	-0.23
02	23100-23256-B	1.0000	2.5625	0.8750	0.4063	0.8450	0.6250	0.06	2.7147	0.1563	1.54	1.36	2.48	0.80	88100	-0.09
03 *	41126-41286-B	1.1250	2.8593	0.9688	0.5000	0.9550	0.6875	0.06	3.0741	0.2187	1.63	1.45	2.76	0.97	11200	-0.16
04	43125-43312-B	1.2500	3.1250	1.0000	0.5313	0.9478	0.6875	0.06	3.3390	0.2188	1.73	1.63	3.03	0.08	12500	-0.08
05	43131-43312-B	1.3125	3.1250	1.0000	0.5313	0.9478	0.6875	0.14	3.3390	0.2188	2.01	1.65	3.03	0.08	12500	-0.08
06	43132-43312-B	1.3125	3.1250	1.0000	0.5313	0.9478	0.6875	0.08	3.3390	0.2188	1.89	1.66	3.03	0.08	12500	-0.08
07	44143-44348-B	1.4375	3.4843	1.0000	0.5312	0.9330	0.6875	0.09	3.6983	0.2187	2.13	1.97	3.39	0.09	13600	0.09
08	44162-44348-B	1.6250	3.4843	1.0000	0.5312	0.9330	0.6875	0.09	3.6983	0.2187	2.24	2.00	3.39	0.09	13600	0.09
09 *	53176-53387-B	1.7500	3.8750	1.2188	0.6563	1.1142	0.8125	0.05	4.1210	0.2500	2.32	2.08	3.62	0.79	16200	-0.01
10	55209C-55437-B	2.0000	4.3750	1.1875	0.5938	1.0594	0.8125	0.14	4.5938	0.2188	2.80	2.54	4.21	0.03	20600	0.30
11	55206-55437-B	2.0625	4.3750	1.1875	0.5938	1.0594	0.8125	0.14	4.5938	0.2188	2.83	2.52	4.21	0.03	20600	0.28
12	55206C-55437-B	2.0625	4.3750	1.1875	0.5938	1.0594	0.8125	0.14	4.5938	0.2188	2.83	2.54	4.21	0.03	20600	0.30
13	66225-66462-B	2.2500	4.6250	1.3125	0.6250	1.2500	0.9375	0.14	4.8750	0.2500	2.99	2.71	4.45	0.93	24000	-0.01
14 *	HM911249-HM911210-B	2.4375	5.1250	1.4375	0.7500	1.3125	0.9375	0.14	5.3750	0.2500	3.35	2.93	4.87	0.71	26900	0.21
15	9380-9321-B	3.0000	6.7500	1.9375	1.0000	1.8125	1.2500	0.14	7.0620	0.3125	4.13	3.87	6.46	0.76	49100	0.17
16	9386H-9321-B	3.3125	6.7500	1.9375	1.0000	1.8125	1.2500	0.14	7.0620	0.3125	4.37	3.87	6.46	0.76	49100	0.17
17	98400-98788-B	4.0000	7.8740	2.0772	1.0772	1.9375	1.3750	0.14	8.2500	0.3750	5.04	4.76	7.40	0.92	65200	0.05
18	37425-37625-B	4.2500	6.2500	0.9063	0.4688	0.8440	0.6250	0.14	6.4336	0.1875	4.80	4.53	6.02	0.54	18900	0.54
19	37431-37625-B	4.3125	6.2500	0.9063	0.4688	0.8440	0.6250	0.14	6.4336	0.1875	4.84	4.57	6.02	0.54	18900	0.54

* INACTIVE FOR NEW DESIGN AFTER 20 MAR 1978.

1/ PART NUMBERS ARE FOR REFERENCE ONLY. PART NUMBERS REFLECT THE CUP AND CONE NUMBERS USED BY INDUSTRY AND THE AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA).
2/ THESE MAXIMUM FILLET RADII SHALL BE CLEARED BY THE BEARING CORNERS.
3/ MINUS VALUE INDICATES LOAD CENTER INSIDE CONE BACKFACE.

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REQUIREMENTS:

1. Material: Cones, cups, and rollers: bearing quality - carburizing grade alloy or through-hardening grade alloy steel in accordance with ASTM-A295, ASTM-A534, or ASTM-A535. Cage: carbon steel (one piece stamped).

2. Hardness: Cones, cups and rollers: 58 to 641HRC.

3. Tolerances: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (d)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0 3.0000	3.0000 6.0000	5 10	0 0

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000 12.0000	12.0000 24.0000	10 20	0 0

Standout (T1)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0 4.0000	4.0000 6.0000	80 140	0 100

Cup flange OD (D1)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	12.0000	20	0

Assembled bearing maximum radial runout		
Cup O.D. (D)		Tolerance
Over	Incl.	
0	24.0	20

4. Effective load center: Dimension (o) locates a point on the cone axis which is the center of pressure of all resisting forces set up by the rollers. All moments should be calculated from this point when determining bearing loading and shaft stresses. A plus value of (a) indicates that the center is outside the cone backface.

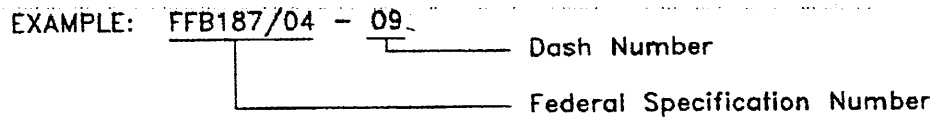
5. Operating temperature: Recommended operating Temperature not to exceed 230° F.

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6. Lubrication: Bearings shall be furnished without lubrication. All bearings shall be preserved in accordance with MIL-P-147.

7. Steep angle: A steep angle bearing has a contact angle (a) between 22 and 31 degrees. The contact angle is the angle between the line of action of the rcller load and a plane perpendicular to the bearing axis.

8. Part number: The part number consist of the letters FFB, the basic federal specification number, the specification sheet number and a dash number from table L



9. For design feature purposes, this standard procurement documents referenced herein.

10. Referenced documents shall be of the issue in effect on the date of invitation for bids, or request for proposal.

NOTE: Inch units can be converted to soft metric units as follows:

Inch	X	25.4	=	millimeters

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INTERCHANGEABILITY TABLE

MILITARY STD PART NUMBERS	SUBSTITUTIVE PART NUMBERS
MS19084-1	FFB187/04-01
MS19084-2	FFB187/04-02
MS19084-3	FFB187/04-03
MS19084-4	FFB187/04-06
MS19084-5	FFB187/04-08
MS19084-6	FFB187/04-09
MS19084-7	FFB187/04-11
MS19084-8	FFB187/04-13
MS19084-9	FFB187/04-14
MS19084-10	FFB187/04-17
MS19084-	FFB187/04-

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